

Beyond the Line of Departure: Cyber Defense Beyond the Perimeter

Modern cyber defense demands AI-driven resilience and a shared fate between cyber defenders and network owners. Previously, cybersecurity teams focused on concepts such as “identity is the new perimeter” and allowed passive responsibility to exist between network owners and cybersecurity teams. Progress in cybersecurity depends on actively defending beyond boundaries and perimeters—not simply standing watch like a Soldier at the DMZ or equating success with avoiding failure. Security teams must integrate graph-based thinking to model IT ecosystem relationships, build complex cyber terrain, employ collaborative intelligence, measure success by detection agility, and adopt a shared-fate model between network owners and defenders to fundamentally change the physics of defense against adversaries.

Rewriting the Rules of Cyber Engagement

Historically, in warfare, defender’s built perimeter-based walls—such as the Maginot Line, the Atlantic Wall, and the Berlin Wall—to force attackers onto unfavorable ground. This approach handed the strategic imperative to adversaries: “the bomber always gets through.” These examples show how operational circumvention made static, two-dimensional barriers of the past fail to account for the third dimension. When defenders rely on static perimeters, isolated log databases, and reactive pass/fail metrics, they enable threat actors to view environments as graph networks. Modern attackers trace credentials, dependencies, and entitlements to find lateral paths towards cyber key terrain and high-value targets (Lambert, 2025). As software complexity, supply chain, and artificial intelligence vulnerabilities increase, defenders who rely on outdated thinking fail and remain locked in a losing conflict (Aubakirova et al., 2025).

Cybersecurity teams must shape the operational environment in which adversaries operate. Cybersecurity through just tabular data logs, or thinking prevention alone is what matters, is the old-guard mantra; defenders must change with the times and adopt the perspective threat actors use (Lambert, 2025). By modeling entire IT ecosystems as connected directed graphs, security teams can predict lateral movement paths, remove bad credential combinations, and actively build “difficult terrain” (Lambert, 2025). When defenders combine multi-directional business and government threat sharing (Lambert, 2025) with AI-driven telemetry mapping (Aubakirova et al., 2025), they incrementally increase the cost and complexity of attackers' tactics, techniques, and procedures.

The Myth of the Identity Perimeter

The popular saying “Identity is the new perimeter” has led cybersecurity teams to believe that authentication alone defends the perimeter. Kenneth Hartman emphasizes that access control relies on a fundamental model that separates accounts into subjects, actions, and objects — “Who can do what to what?”—that is, authentication and authorization (Hartman, 2025). Identity represents the principal making a request, while the perimeter provides the physical and/or virtual boundary where verification, policy enforcement, and governance occur. Identities cross boundaries; they do not form the boundary itself, so saying identity replaces the perimeter is incorrect (Hartman, 2025).

Under a strong Zero Trust framework, identity operates across localized micro-perimeters (Hartman, 2025). Each interaction crosses a potential trust boundary in cloud environments. Security teams create micro-security perimeters throughout the

environment by strictly micro segmenting—separating individual workloads within Virtual Private Clouds (VPCs) or Virtual Networks and restricting inter-service communication to only the minimum necessary ports and sources (Hartman, 2025), like individual fuses in a circuit breaker. Governance and policy enforcement rely on this distributed segmentation.

Measuring Success Beyond Failure

Cybersecurity teams must replace binary pass/fail security metrics with probabilistic, continuous metrics (Lackey, 2019). The old guard in IT relies on indicators like uptime or mean time to repair, and cybersecurity teams have traditionally assumed that if no breach is known to have occurred, security has succeeded (Lackey, 2019). “The default assumption of integrity, where absence of evidence is routinely mistaken for evidence of absence, is a posture of hope, not a defensible strategy” (Szewczyk, 2025). Because no perimeter is infallible, teams should assume attackers are already on the network by default (Lackey, 2019).

When organizations assess security based solely on avoiding failure, they miss opportunities for gradual improvement and remain blind when attackers bypass perimeters (Lackey, 2019). Teams achieve gradual improvement by focusing on visibility and detection speed—using metrics such as mean time to detect and mean time to respond, or detection velocity. Cybersecurity teams should assess security by asking three operational questions (Lackey, 2019):

1. What are the most likely attacker courses of action?
2. How fast can we detect the attacker?

3. How can we make these courses of action gradually more difficult and expensive for adversaries?

Shifting the Physics of Defense & Creating Difficult Terrain

Graph-Based Thinking vs Relational Tabular Data

For defenders to gain the strategic imperative, security teams must shape the operational conditions under which attackers operate. Presently, security operations heavily rely on tabular log data to infer relationships; this obscures attackers' probabilistic courses of action (Lambert, 2025). Attackers in cyberspace view targets as channeled graphs of credentials, entitlements, and software dependencies, searching for lateral paths towards key terrain and high-value targets (Lambert, 2025).

To adapt, cybersecurity teams must adopt graph-based thinking and build attacker courses of action that trace activity across different telemetry sources (Lambert, 2025). Defenders and AI systems can use what Lambert calls “algebras of defense”—relational tables, attack graphs, anomaly models, and temporal vectors—to show blast radii, identify weak links, and bad credential combinations, and remove potential lateral movement paths before attackers use them (Lambert, 2025).

Creating “Difficult Terrain” & Deception

After graph-based models, defenders must build “difficult terrain” to increase adversary effort and financial cost (Lambert, 2025). Difficult terrain or preventative hygiene could encompass:

- Sunsetting legacy systems and older protocols.
- Regular removal of unused accounts, permissions, and abandoned resources.
- Enforcing phishing-resistant multifactor authentication (MFA).

- Ensuring administrative access is only granted to hardened network locations.

Defenders using deception - such as canary tokens and modern honeypots- help create difficult terrain, increase the signal-to-noise ratio, destroy attackers' predictability advantage, and force them to make high-fidelity mistakes. No defense is perfect; organizations that are the most collaborative and trusting, have a collective mindset, share real-time threat data and telemetry regarding cyber intelligence thrive.

Breaking the Cyber Kill Chain with AI

Cybersecurity now suffers from tool sprawl, with dozens of vendor-specific tool terms. Jargon distracts from understanding tools' main purposes (Aubakirova et al., 2025). Security leaders can maximize impact by evaluating how effectively tools disrupt the Cyber Kill Chain. Overlapping acronyms such as SIEM, SOAR, EDR, CNAPP, DSPM, and XDR describe vendor categories, not specific attack stages, which should be the main focus (Aubakirova et al., 2025). Superior tools simplify and focus on how they intervene in an intrusion and how effectively they detect and respond to threats.

AI Asymmetry: Left-of-Boom vs Right-of-Boom

Shifting from a reactive to proactive posture in cybersecurity requires the use of Artificial Intelligence to counter machine-speed reconnaissance and surveillance with machine-speed containment and intelligence (Aubakirova et al., 2025). Attackers already utilize AI as an offensive force multiplier to generate complex phishing attacks, mutate malware signatures, and scan at scale (Aubakirova et al., 2025). AI for defense must be operationalized on both sides of the “boom.

Left-of-Boom (Prevention & Delivery): Using AI can help build anomaly baselines from user activity, network traffic, and devices; these can indicate reconnaissance and phishing attempts that bypass signature filters (Aubakirova et al., 2025).

Right-of-Boom (Command and Control & Lateral Movement): AI correlation engines and large language models (LLMs) put together separate, low-fidelity signals like an isolated login anomaly, suspicious DNS queries, or a sudden privilege escalation - into coherent kill chains (Aubakirova et al., 2025). Instead of being burdened with thousands of siloed alerts, LLMs can capture context for the defender (Aubakirova et al., 2025).

Defenders can use AI to provide context and correlation, proactively detect and respond to threats, and change the economics and physics of cyber defense in their favor (Aubakirova et al., 2025).

Operationalizing Shared Fate in Cloud Security

Current policy and security governance lag behind the changing operational environment. The old-guard model of Shared Responsibility, which clearly defines security tasks between network owners and defenders, dominates most frameworks. Evidence demonstrates that the traditional Shared Responsibility model fails to achieve better security outcomes (Google Cloud, 2023). Shared Responsibility encourages passivity; network owners must figure out thousands of evolving configuration settings across different services and devices (Google Cloud, 2023). Human error, policy or governance gaps, and complexity cause misconfigurations—the main reason attackers succeed (Google Cloud, 2023).

Transitioning to Shared Fate

The new Shared Fate model between network owners and cybersecurity teams corrects the flaws in the shared responsibility framework (Google Cloud, 2023). Shared Fate requires an active, ongoing partnership between the network owner and the cybersecurity team who pursue the same operational goal: eliminating risk (Google Cloud, 2023). Security architects should design infrastructure from scratch with input from the cybersecurity team; shared accountability demands opinionated design and security-by-default for infrastructure, policies, and governance (Google Cloud, 2023). Teams that embed security and configuration directly into the infrastructure using secure blueprints form an essential part of the secure-by-default strategy (Google Cloud, 2023).

Shared Fate works best when teams integrate it into policy, governance, and risk-sharing programs (Google Cloud, 2023). Network owners should support continuous posture management with native policy enforcement tools and AI-driven automated threat detection. This approach reduces risk, so the network and cybersecurity teams share a common destiny.

The Way Forward

Passive compliance exercises and defending static perimeters no longer define effective cyber defense. As the cyber operational environment and terrain grow in complexity and speed, defenders must reject simplistic boundary-oriented defenses and pass/fail metrics. To achieve progress, cybersecurity teams must become agile and dynamic, consistently choosing approaches that force adversaries onto unfavorable ground at every stage of the attack lifecycle.

Teams change the physics of the operational environment by using the “algebras of defense”—adopting graph-based thinking to visualize connections and anomalies over time. Security leaders must leverage collaborative intelligence, create difficult terrain (including the use of deception), and integrate AI-native analytics to accelerate detection both left and right of “boom.” Teams must also adopt the Shared Fate framework because, as technology becomes obsolete, collaboration and human capital remain the enduring advantages.

Author Note

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